

Institute of Actuaries of India

Subject SA3 – General Insurance

May 2014 Examinations

INDICATIVE SOLUTIONS

Solution 1 :

i. Items to consider in quantifying the tail assumptions:

Length of Tail: The construction projects last up to 8 years. There may still be claims and recoveries yet to be finalised by the end of the contract term. Hence the tail assumptions need to capture incurred developments well after year 6.

Pattern of exposure: What is the pattern of exposure from the various engineering projects? Is the pattern a decreasing one over time, or is there a “spike” in claims incurred toward the end of the 8 year period.

Nature of claims in the tail: Types of claims that are likely to be exposed here relate to non-completion of projects or faulty design. These claims, if they occur, are likely to be material in relation to the overall claims cost, so it is important to allow for them.

Information sources to search out in quantifying the tail assumptions:

- Given that Alternative is a relatively new insurer, it will not have data on these tail claims as yet. So, other information sources may need to be looked into, e.g industry experience or behaviour of similar portfolios if they exist, consultants or relevant reinsurers
- Discussions with the claims staff to understand how they have set the case estimates and therefore to understand the robustness of these estimates and the potential for any deterioration or redundancy.
- A review of the length of contracts for each of the underwriting years – naturally years with a shorter average contract term will be expected to have lighter tail experience than those with longer average terms.
- Discussions with underwriters and engineers who underwrote the various projects may shed some light on potential quantum of losses and their likelihood of occurrence.

[4]

ii. a) Deriving the Ultimate Loss Ratio and the Projected Future Claims Payments for the 2012 underwriting year.

Calculating the cumulative factors

	1:2	2:3	3:4	4:5	5:6	6:7	7:8
Paid CL factors	2.07	1.25	1.13	1.08	1.04	1.02	1.08
Weighted average	3.62	1.75	1.40	1.24	1.15	1.10	1.08
Cumulative factor							

Calculating the 2012 Underwriting Year Ultimate Incurred Loss Ratio

Paid To Date	2,421 (A)
Calculated CL Factor to Ultimate	1.750 (B)

Deriving the 2012 Underwriting year IBNR amount

Ultimate Incurred Estimate	4,237 (C) = (A x B)
Written Premium	22,791 (D)
Ultimate Loss Ratio	18.6 % (E) = (C) / (D)
Ultimate Incurred Estimate	4,237 (E)
Paid To Date	2,421 (A)
Case Estimates	556 (F)
IBNR Amount	1,260 (G) = (E) - (A) - (F)

[3]

b. Selection of Assumptions

	Tail Factor Assumption						
	1:2	2:3	3:4	4:5	5:6	6:7	7+
Weighted Average LR increase	10%	7%	5%	4%	2%		
Selected LR increase	10%	7%	5%	4%	2%	1%	5%
Cumulative LR increase	34%	24%	17%	12%	8%	6%	5%

Calculating the 2013 Underwriting Year Ultimate Incurred Loss Ratio

2013 Incurred Loss Ratio	22% (A)
Cumulative LR increase for 2013	34% (B)
Ultimate Incurred Loss Ratio Estimate	56% (C) = (A + B)

Deriving the 2013 Underwriting Year IBNR amount

2013 Written Premium	21,843 (D)
Ultimate Incurred Estimate	12,232 (E) = (D) x (C)
2005 Incurred To Date	4,723 (F)
IBNR Amount	7,509 (G) = (E) - (F)

[3]

iii. Issues to consider when re-selecting your assumptions for the additive model to allow for the change in rates. This is a challenging task. For the historical loss ratios to be good predictors of future experience, the expected ultimate loss ratio for each underwriting year must be in a similar range. The doubling of rates means that the ultimate loss ratio for the most recent underwriting years will be halved (if we assume claims incurred are unchanged). Indeed, there is implicit evidence in the data of lower loss ratios for the 2011 to 2013 underwriting years compared to prior underwriting years at the same stage of development.

To allow for this change, a number of considerations need to be taken into account. For example, selecting two sets of additive Loss Development Factors (LDFs), one for pre and one for post rate change. The LDFs for the recent underwriting years will be lower than those for older underwriting years in proportion to the premium rate change.

Another aspect is the impact of the rate change on the types of risks underwritten. Given the doubling of rates, has the mix of risks changed? E.g. different size of projects, different project durations etc. If so this may entail a different pattern of incurred development.

Selection effect should also be allowed for, as the 'good' risks may choose not to insure as much as before the rate change (e.g. by self-insuring or retaining a greater part of the risk), while the 'bad' risks may still view the insurance price as being more 'value for money' compared to the 'good risks'. The selection effect means that the doubling of rates will not necessarily lead to halving of the loss ratios.

Need to also look into the reason why rates were doubled, so far (in the above) we have assumed that claims incurred will more or less remain unchanged (barring any selection effects). If the rate doubling is due to expected claims cost increases e.g. higher building rectification costs or higher expected inflation or some developments that will affect claims costs going forward (which have not shown up in the triangle as yet), then the LDFs should be adjusted upwards according to the estimated expected increase in claims costs

[2]

iv. a) The paid chain ladder method is a multiplicative method (it multiplies a factor by the current paid to date amounts). It assumes that future experience is proportional to that seen to date, which may not hold true if the low payments to date are due to a slowdown in the payment pattern. Weaknesses also arise because the method does not utilise crucial case estimates information, an important feature of the Engineering class in question which has a tail running to 8 years and beyond.

[1]

b) The additive chain ladder method assumes that each year's case estimate development pattern will be similar. If case estimates are underestimated for any reason, then this method will result in an understated liability. Another issue is that the method assumes that the future incurred experience will be proportional to premium, therefore it will be distorted by premium rate changes over time and needs to be adjusted for this.

[1]

v. Additional issues would you need to consider when estimating the outstanding claims provision for the 2012 underwriting year:

- Given the policy covers claims incurred over many years, the premium for the 2012 underwriting year won't have been fully earned by the end of 2013 which is the assumed date of the outstanding claims analysis. You would therefore need to consider the pattern of premium earning such that the projected future claim payments could be divided into 'outstanding claims' and 'premium liability' components.
- The appropriate Claims Handling Expenses to be added into the provision.
- Any changes in underwriting guidelines and processes over time?
- Any changes in claims management and case estimation philosophy over time?
- Any large claims information that may influence your opinion about the best estimate?
- Which projects have a long duration (6-8 years) as they will influence the tail
- What are the reinsurance covers that protect Alternative, to derive the net OCP?
- Need to perform a sensitivity testing of the final result
- Previous actuarial reports that may shed some light on the issues above.

[4]

[Total Marks-18]

Solution 2 :**i. Exposure information :**

Issues identified by underwriter:

- Rating by profession and assessment of riskiness.
- Schemes.
- Exposure measure, e.g. turnover – will want to classify insureds into broad groups based on size – small/medium/large.
- Identification of reduction in exposure corresponding to reduction in premium income.
- Growth – identification of hardening rates corresponding to risk assessment.

Request data for each underwriting year (or if accident year, needs to be apportioned on a sensible earning basis from underwriting year) split into schemes/ordinary:

- Number of policies
- Annual turnover or other exposure measure (but not just premiums as a proxy)
- Premium income
- Rate charts for the relevant period, and if possible any competitor's rate charts.

We need exposure data to test the assertion of decreasing relative riskiness of professions.

We want the rate charts to examine loadings/discounts on the insurer's base rates and the comparison to competitors would be helpful.

For each scheme:

- Membership profile summary (unlikely to get detailed individual statistics, but would like some small/medium/large splits if relevant)
- External data about scheme coverage of profession – are there any particular biases in the profile of this scheme relative to the profession as a whole – e.g. it only tends to attract large corporate players.

Underwriter feedback

- Which professions were riskier and why and which professions are now being targetted and why?

[4]

ii. Claims data :

Issues identified:

- Schemes versus ordinary
- Change in mix of business i.e. relative riskiness of professions
- Change in finalisation rate
- Change in case estimation

Need to get separate claims triangles for schemes and ordinary business split by profession – specify corresponding accident year or underwriting year to (a).

- Number of claims reported
- Number of claims finalised
- Number of claims reopened
- Number of claims finalised for zero payment

- Claims payment
- Case estimates (to get claims incurred triangles)

Will also want this triangle data for large claims (usually based on an incurred figure and with reference to reinsurance retentions). Also will want an individual claim listing of these large claims and some narrative description.

Will want reinsurance arrangements for the period – retentions, indexation clauses etc

For schemes – most likely recorded as single ‘claims’ in the insurer’s system, will want to get underlying claim statistics from the broker.

Given the change from 3-year to 1-year activity would like to get a separate claim triangle for the past zero finalisations and reporting so that it can be recast to assess it on the new basis.

Given the change in case estimation, would like some idea of what the increase in average case estimate is. This is unlikely to be available for all claims and may just request it for the large claims – there will be fewer of them. Also if possible would like to get separate payment and case estimate histories for these large claims – it will then be possible to test the assertion about savings on case estimates.

An alternative would be to get quarterly development triangles as a means of testing the continuous estimation procedures and the zero-finalisations.

Note that while we would like information split by profession, we may well have to aggregate it up into groups of professions because the data will otherwise be too sparse for meaningful analysis.

[5]

iii. Outstanding claims estimation techniques :

Issues:

No single method is likely to suffice, given all the changes going on.

Paid case estimates – unlikely to be reliable for recent years because of changes in case estimation, but may be appropriate for older years.

Average cost per claim finalised – natural model choice for PI business given that the majority of any settlement is made at finalisation. Problem here is the change in the finalisation pattern. If additional data is requested, as suggested above, then it will be possible to make a sensible back-cast of data to come up with a sensible finalisation pattern. However, if the proportion of non-zero finalisations is not steady (and it may not be if the change in exposure has occurred) then this will increase the uncertainty of the method.

Average cost per claim in operational time – an alternative to the standard average cost per claim finalised method. The change in finalisation pattern may be able to be more directly addressed using this method. Separation of zero- and non-zero claims may be necessary to preserve invariant order assumption.

Average cost per claim incurred – this method should probably be mentioned as one means of avoiding the problems associated with the change in finalisation pattern. The problem is that the change in exposure basis will mean that this method may not be fully reliable either.

Bornheutter Fergusson – this method requires judgmental input regarding starting loss ratios and will also require some adjustment for more recent changes to case estimation procedures, but it should also be considered

All of these methods should be considered for data split by scheme/non-scheme and potentially for individual professions – although some aggregation is likely to be needed.

The analyses should be carried out on ‘small’ claims and a separate allowance for large claims needs to be made. The estimate of large claim frequency (per dollar of exposure or per thousand reported claims or some other measure) will be affected by the change in case estimation and if the additional data is available then some sensible reduction to the apparent frequency will be possible.

Overall, the results from each of the methods should be compared and apparent anomalies investigated. A final central estimate is likely to be based on an average of methods or some other amalgam.

[8]

[Total Marks-17]

Solution 3 :

i. Value at Risk can be used to calculate an exceedence probability. This is the probability that the actual loss will exceed the expected loss plus allocated capital.

To do this, get the distributions which describe each line of business losses. Select a % probability of ruin that is relevant. Let's say 1%. Calculate the 99% VaR for each line of business and allocate proportionally to that.

Determine the value of loss associated with each percentage. Determine an aggregate (all line) distribution. Determine the proportion of each line's value to the aggregate.

Issues to consider

Value at Risk does not consider the severity of losses exceeding the threshold.

There may not be enough capital to allocate so all lines have the target exceedence probability

Diversification benefits between lines of business are not considered

It is important to select an appropriate time horizon because assets and liabilities change with time and so does risk capital. Should select a short time horizon as VaR does not work well over long time periods because the tail of the normal distribution is too thin and data will be insufficient for empirical analysis

[4]

ii. VaR is simple to understand and easy to calculate

VaR allows for same %measure to be used across lines of business so as to provide an easy comparison among risks/capital needs of lines in an equivalent manner.

VaR enables capital to be allocated to various lines based on the risk level at that threshold. VaR is probability based so it can be used to develop a result at any level of confidence

Since VaR is based on the normal distribution, calculations are understandable and straightforward

[4]

iii. The risk factors to be considered in projecting the financial and capital adequacy positions over a three year time horizon are:

- Claim rates: Yearly deterioration in experience over the projection period
- Investment returns :

- (a) Yield curve
 - Parallel yearly yield curve shift
 - No change for duration less than 5 years, yearly parallel shift for duration more than 5 years, linear interpolation
 - Yearly parallel shift for duration less than 5 years, no change for duration more than 5 years, linear interpolation
 - Change in credit spreads
 - (b) Equity market
 - Equity total returns deteriorate over the projection period
- Expenses and renewal rates: Yearly deterioration in experience over the projection period
 - New business: Yearly fall in new business income over the projection period
 - Where material, the Appointed Actuary should include scenarios on deterioration of the company's experience over the projection period arising from changes in:
 - Reinsurance ceded
 - Reserving basis
 - Taxation

[7]

[Total Marks-15]

Solution 4 :

i. Extreme value theory (EVT) studies probabilistic models for the occurrence of rare events. In the context of general insurance extreme events can be defined as very unlikely or rare events that have a large financial impact.

Some examples of extreme events include:

- Natural catastrophes
- Manmade catastrophes
- Financial events

Within general insurance companies, extreme value theory is often used in:

- Pricing: *eg* deriving expected loss costs for excess of loss reinsurance contracts for either high layers or for covered perils which are rare events.
- Reserving: *eg* deriving the expected recoveries from excess of loss protections of pure IBNR losses where the protections are either high layers or for covered perils which are rare events.
- Capital Modelling: *eg* deriving tail distributions for large or catastrophic losses.

The two key extreme value distributions that are used in general insurance application are: GEV distribution and the Generalised Pareto Distribution.

[4]

ii.

- Proportional Reinsurance – quota share
- Stop loss cover
- Per risk excess of loss cover

[2]

iii.

a. Insurer:

1. Advantages

- a. Possible lower retention levels as compared to quota share
- b. Transparent processing as data populated from weather stations
- c. Greater certainty with respect to business underwritten
- d. Can negate the requirement of a further catastrophe XoL reinsurance
- e. Technical help from reinsurers

2. Disadvantages

- a. No ceding commissions
- b. Cost of purchase can be a sunk cost specially for very high return periods, can get redundant if the Company already has an whole account CAT XoL
- c. May not be very useful from capital management perspective (unlike proportional that can provide immediate capital relief)

b. Reinsurer:

1. Advantages:

- a. Greater access to claims and historical data of the insurer
- b. Can have say in underwriting policies too considering the importance of setting triggers
- c. Transparent claim processing If the return periods are very high, can be a source of near certain income
- d. For high return periods, the treaty can operate at zero loss ratios.

2. Disadvantages:

- a. Greater uncertainty with respect to risks accepted
- b. Potentially low ceded premiums vis-à-vis proportional reinsurances
- c. Can be difficult to find reinsurers for retrocession for such tailor-made covers, so high capital strain for reinsurer

[6]

iv. Comments:

This would need wider understanding of the concept.

- Probability weighted moments
- Bayesian methods
- Bootstrapping methods
- Method of moments

[2]

v. Comments:

Requires wider / higher order understanding

- a. Histogram and plotting the empirical cumulative distribution function
- b. Q-Q plot : to see if the parametric model fits data well
- c. P – P plot: akin to Q-Q plot above
- d. Quantile-Quantile plot of residuals of a fitted model
- e. Gumbel/Weibull/Frechet plot: to evaluate which model suits best
- f. Mean excess plot
- g. Time series plot: to assess seasonality, cyclical behavior, trend estimation
- h. Autocorrelation plot: whether time-dependent correlation is zero (random) or non-zero (non-random)

[8]

vi. Comments:

Requires conceptual understanding of the subject matter that needs mathematical application

The location parameter explains the shift of the distribution, i.e. it translates the graph by shifting it to the right or to the left based on the magnitude of the shift.

The scale parameter typically explains the dispersion in the distributions i.e. whether the distribution is expanded or condensed.

The shape parameter explains the tail of the distribution i.e. the effect of outliers or extreme values of the distribution

Location parameter = $\mu = 6.0468$

Scale parameter = $\sigma = 0.8635$

Shape parameter = $\xi = -0.5728$

[4]

vii. Substituting values for μ , ξ and σ in the CDF provided for $\xi \neq 0$

$$\begin{aligned}
 1. \exp \left(- \left(1 + \xi * (x-\mu)/\sigma \right)^{-1/\xi} \right) &= 1 - 1/10 \text{ (return period 10 years)} = 0.90 \\
 - \left(1 + \xi * (x-\mu)/\sigma \right)^{-1/\xi} &= \ln(0.9) = -0.10536 \\
 (x-\mu)/\sigma &= 1.264764 \\
 x &= 7.14
 \end{aligned}$$

$$\begin{aligned}
 2. \exp \left(- \left(1 + \xi * (x-\mu)/\sigma \right)^{-1/\xi} \right) &= 1 - 1/25 \text{ (return period 25 years)} = 0.96 \\
 - \left(1 + \xi * (x-\mu)/\sigma \right)^{-1/\xi} &= \ln(0.96) = -0.04082 \\
 (x-\mu)/\sigma &= 1.466352 \\
 x &= 7.31
 \end{aligned}$$

$$\begin{aligned}
3. \exp(- (1 + \xi * (x-\mu)/\sigma)^{-1/\xi}) &= 1 - 1/50 \text{ (return period 50 years)} = 0.98 \\
- (1 + \xi * (x-\mu)/\sigma)^{-1/\xi} &= \ln(0.98) = -0.0202 \\
(x-\mu)/\sigma &= 1.559028 \\
x &= 7.39
\end{aligned}$$

$$\begin{aligned}
4. \exp(- (1 + \xi * (x-\mu)/\sigma)^{-1/\xi}) &= 1 - 1/100 \text{ (return period 100 years)} = 0.99 \\
- (1 + \xi * (x-\mu)/\sigma)^{-1/\xi} &= \ln(0.99) = -0.01005 \\
(x-\mu)/\sigma &= 1.620598 \\
x &= 7.45
\end{aligned}$$

$$5. \text{Upper bound given by } \mu - \sigma / \xi = 7.55$$

[11]

viii. A negative shape parameter is characteristic of thin tailed distributions or distributions with an upper bound. However, with rainfall measures, it may not be appropriate to assume an upper boundary.

Besides, the model appears to be an inappropriate fit since the maximum observed rainfall is 7.7 cms which is significantly lower than the upper bound trigger of 7.55 cms.

A good model should also be able to explain the outliers with a non-zero probability without completely excluding them.

The reinsurance arrangement does appear inappropriate considering the very small variation in trigger levels amongst the various return periods. For a reinsurance arrangement of this kind, it is imperative that the dispersion in the extreme values (above a certain threshold) is significantly large as, otherwise the cost of purchasing the cover can be exorbitantly high for the insurer.

The extreme value theory models the distribution of the extreme values. But from the results above, one may have to infer that the given fit to data shows a very thin or virtually non-existent tail, which may not be suitable for extreme value modeling assumptions.

Also the shape parameter being negative for July to September period (characterized by monsoons in India) does not appear to be a valid fit.

[6]

ix. Model fitting:

Further factors / next steps:

1. Other models like Generalized Pareto Distribution at a pre-fixed threshold trigger point
2. Re-model using the GEV under different set of assumptions using other distribution that have a positive shape parameter (no upper bound)
3. Whether all data was historical or was there a presence of extrapolated data
4. Revisit *a priori* assumptions
5. To rerun the exploratory analyses
6. To investigate if there were errors in the data – possibly use data for longer periods
7. To assess if the rainfall data exhibits mean reversion properties as that could invalidate the EVT argument as the variable may no longer satisfy the i.i.d property
8. Estimate parameters using other methods
9. Use a frequency distribution of rainfall index above a given threshold

Reinsurance:

Other factors to be considered / next steps:

Simultaneous to improving the EVT model, the Company may consider other reinsurance arrangements

10. Was there a parameter error leading to size of the tail being underestimated? Subjectively change the scale parameter and run scenarios
11. Effect of inflation on retention level – indexation / stability clause
12. Hours / days clause e.g. minimum three days of rainfall exceeding the trigger or any such definitive clause for risk
13. Traditional catastrophe reinsurance cover
14. Regulatory implications of such an arrangement
15. Run the profit model and capital model in conjunction with other reinsurance options
16. Reinstatements, if any

[7]

[Total Marks-50]
